From Interconnect Materials and Processes to Chip Level Performance: Modeling and Design for Conventional and Exploratory Concepts

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2020 IEEE International Electron Devices Meeting (IEDM) | 978-1-7281-8888-1/20/$31.00 ©2020 IEEE | DOI: 10.1109/IEDM13553.2020.9371945

***Abstract*—**We survey latest device and interconnect scaling trends in literature and present an in-detail sensitivity analysis of advanced metallization and barrier/liner fabrication methods at a system level by evaluating representative designs using industry-standard electronic design tools. In particular, we look at the 7nm technology node and evaluate multiple process advancements for copper interconnects and vias and compare them with ruthenium wires and interconnects. Despite improvements to copper barrier/liner, vias, and geometry RC optimization, by utilizing system-level design metrics, we showcase that ruthenium is a viable option even at the 7nm node, sooner than projections made in literature, owing largely to the 5× improvement in ruthenium via resistance.

I. Introduction

Relentless scaling of transistors and wires to enhance transistor density and system-level performance has resulted in severe implications on advanced metallization processes and performance. Back End of Line (BEOL) parasitics are increasingly dominated by interconnect resistance due to severe size effects in copper wires and challenges to metal barrier scaling [1, 2]. New materials have been explored to replace copper wires and vias, like cobalt and ruthenium, that exhibit higher bulk resistivity (ρ_{bulk}) values compared to copper but can lead to lower effective resistivities due to thin barrier-liner requirements and lower size effects. Literature has shown that the crossover point for ruthenium resistance is beyond the 7nm node, when wires/vias are scaled to around 14nm critical-dimension (CD) [3]. In this paper we (i) review the scaling trends for R and C values of interconnects and transistors, (ii) perform a detailed sensitivity study focused on the impact of BEOL processes on system-level performance, (iii) evaluate copper technology at the 7nm node at a system level with various enhancements to barrier/liner, vias and Aspect-Ratio (AR) optimizations to push copper interconnect performance/power, and (iv) compare the flavors of 7nm copper interconnect based systems with designs based on ruthenium wires and vias and showcase the crossover point where it is beneficial to replace copper wires/vias.

II. Device and Interconnect Scaling Trends

Device and interconnect technologies defined and projected for a wide spectrum of technology nodes, from 32nm to 3nm, are surveyed [4-8], and the device RC is compared against the interconnect RC by assuming a wire length of 100 Metal-Pitch (MP) (Fig 1-2). With per-unit-length interconnect capacitance remaining relatively fixed across technology nodes, dimensional scaling leads to shorter local and intermediate interconnects and less capacitance for the same design specifications (this may however change if new systems are designed with larger memories and floorplans). At the same time, device capacitance has increased with the transition from bulk/planar devices to FinFET devices, as shown in Fig. 1. We observe from the ratio of device to interconnect capacitance that interconnect capacitance becomes less of a factor at advanced nodes (Fig. 1); another factor leading to this conclusion is the interconnect resistance rapidly increasing at scaled dimensions. We observe an overall decrease in device resistance with advancing technology, with a noticeably larger drop in resistance when transitioning from bulk to FinFET technologies which was done to achieve better electrostatic control and higher current drive (Fig. 2). When we compare the interconnect resistance with device resistance, we see interconnect resistance becoming a larger share of the overall path delay. With the capacitance and resistance changes at advanced nodes, there are proposals to improve interconnect resistance by either widening [9] or using higher AR wires at the expense of increasing parasitic capacitance to help alleviate the resistance bottleneck at the local metal levels.

The major limitations of copper technology have come down to the barrier/liner requirements, and the implication of this requirement on wire and via resistances is substantial [12]. Cobalt and ruthenium have shown to provide less severe size effects and become competitive below a CD of 14nm (assuming ultra-scaled liner widths are possible for copper wires), largely due to having lower mean-free-path (λ), lower $\rho_{bulk}*\lambda$ which serves as a proxy for size effects, and higher melting-point/activation energies enabling liner-free/thin liner wires/vias [10-11].

III. Modeling and Benchmarking Methodology

We present the place and route (P&R) results of different interconnect scenarios on power, performance, and area (PPA). We perform the analysis for two representative circuits which have different importance for interconnect and via in design PPA metrics (Fig.3, [12]); the two designs are: Advanced Encryption Standard (AES), which has random logic and moderate routing demand, and Low-Density Parity Check (LDPC), with its high routing demand [13]. The BEOL resistance models are based on published models for Cu interconnects that have been calibrated to experimental data [14-15], and the RC extraction for the BEOL stack is performed using Cadence® Quantus™. The technology is based on the predictive ASAP7 7nm Process Design Kit (PDK) [16] as specified in Table I.

The designs are all evaluated under an ISO-area constraint. We then establish a reference baseline scenario based on current industry's Best-Known Method (BKM) obtained from [17-18]. We assume a copper BEOL stack with a metal barrier

(MB) that has a bottom thickness of 4nm. All other BEOL scenarios defined in Table II are benchmarked against the base BKM case.

IV. Interconnect Scaling and Sensitivity Scenarios and performance Results

The final design implementations are presented in Fig 10. Prior to delving into the detailed sensitivity analysis, we analyzed the delay breakdown for the top critical paths with the BKM case, and we see that the net delay contributes 10% and 35% of the total delay for the AES and LDPC circuits, respectively (Fig. 7). Hence, our expectations for any interconnect process optimizations are limited within these bounds for these designs. Since the AES design is less wire-delay dominated (Fig. 7), we expect the impact of improving interconnect technologies will be more apparent in the LDPC circuit. We attempt to isolate the PPA improvement obtained from just the BEOL process improvements and the additional improvement gained through design tool optimization algorithms by performing one set of analyses by plugging in the new wires/vias in an existing design and extracting PPA (Table III-IV), and the second set of analyses complemented with EDA optimizations (Table V-VI).

A. Scenario 1: BEOL Stack Metal Barrier Scaling

In addition to rapidly increasing interconnect resistance, the highly resistive MB has not scaled at the same rate due to process and reliability concerns. At 18nm trench width, the MB consumes as much as 40% of the wire volume (Fig 6). In the first experiment, we benchmark the impact of thinning the MB material (MB_{thin}) from 4nm to 2nm, which improves both line and via resistances; and in the second, we only improve via resistance (Via^{++}) as in Table II and [8].

Without EDA optimizations, thinning barrier materials (MB_{thin}) and improving vias (Via^{++}) show similar improvements for the AES circuit (Table III). In the interconnect dominant LDPC design, MB_{thin} shows a greater (i.e. 15.56%) improvement in performance compared to 9.19% for Via^{++} highlighting the importance of both wire- and via resistance to system performance (Table IV).

With re-optimization, we see additional performance gains for both MB_{thin} and Via^{++} for both circuits (Table V-VI). The tool extends the performance gains while using less buffers and wiring, which leads to a better power delay product (PDP).

B. Scenario 2:Local Interconnect/Via Metal Barrier Scaling

The wire resistance drastically changes from node-to-node at the local metal levels (Fig. 4-5), hence in the following sections we focus on local metals only. In addition, we perform a sensitivity analysis on the impact of improving only wire resistance (M_{local}-MB_{thin}(wire)) and only improving via resistance (M_{local}-MB_{thin}(via)).

Without EDA optimizations, thinning only the local interconnect MB in the design results in 0.6×-0.8× the improvement achieved by thinning the entire BEOL stack, highlighting that the local metals are largely limiting the wire-delay dominated critical paths (Table III-IV). Additionally, via resistance is more important in improving the critical path delay. When we look at the wire:via resistance for the top critical paths in Fig. 8-9, we see that via resistance is a larger part of the total net resistance for the AES circuit compared to the LDPC circuit. With EDA optimizations, we see the tool compensating for the better wire resistance for the LDPC circuit. M_{local}-MB_{thin}(wire) resulted in a 16.1% improvement compared to 11.6% for the M_{local}-MB_{thin}(via) case (Table V-VI).

C. Scenario 3:Alternative Metals and High Aspect Ratio

We explore new materials and schemes for the local interconnects. The first case looks at the impact of replacing local interconnects with Ruthenium (M_{local}-Ru). Ruthenium as an alternative metal does not require a MB and is less sensitive to size effects compared to Cu. Based on published results [3,19] the line and via resistances are shown in Figs. 4 and 5. At 18nm CD, Ru line resistance already performs better than Cu with 4nm MB. Because Ru doesn't require a resistive MB, the via resistance is superior to copper. We also explore high-AR wires, which have been proposed to alleviate the resistance bottleneck at the expense of parasitic capacitance. We consider high AR interconnects (i.e., AR=3) for both Ru (M_{local}-Ru-AR3) and Cu (M_{local}-BKM-Cu-AR3). From our parasitic extraction for M1-M3, we see AR3 can improve resistance 40% while increasing net capacitance 28%. For cases where resistance is dominant, this tradeoff in resistance and capacitance can help improve overall performance.

From our results (Table III-VI), Ru local metals have similar and sometimes better performance gains than the MB_{thin} case motivating the move to ruthenium sooner. For the high AR cases, we see little change in performance but a larger hit to power due to the increased capacitance.

V. Conclusion

In this paper, we have performed a detailed sensitivity analysis of BEOL processes on advanced node physical design implementations. In particular, we have looked into implications of barrier-liner, via bottom- and side-wall liner thickness advancements for copper wires and explored ruthenium-based wires with geometry optimizations for improved design PPA. With rapidly increasing via resistances and dense local connections, via resistance plays an important role in overall system performance. At the 7nm technology node, Ru interconnects already show good improvement to performance. Ru line resistance is already comparable to Cu and the barrierless vias have much lower resistance compared to Cu, contributing more to the gain in performance. For the high AR Ru and Cu, we see very little performance gains for all cases at the 7nm technology despite better resistance. Power dissipation also increases due to higher interconnect capacitance. However, we expect larger performance gains with high-AR wires at more advanced nodes when resistance is further exacerbated due to dimensional scaling.

Acknowledgment

The authors gratefully acknowledge Divya Prasad and Brian Cline from Arm for their guidance in this research. They also thank Wriddhi Chakraborty for the literature on CMOS scaling trends. This work was supported by ASCENT, one of six centers in JUMP, a Semiconductor Research Corporation (SRC) program sponsored by DARPA.

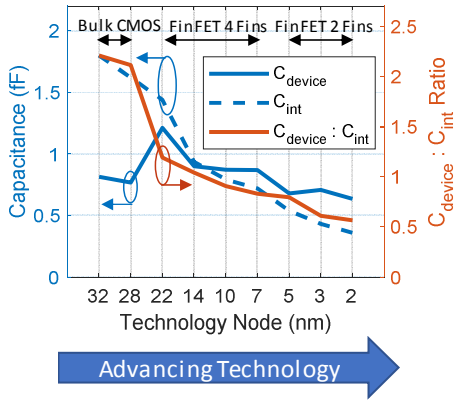


Fig 1. Interconnect and Device Capacitance Trends for advancing technology nodes

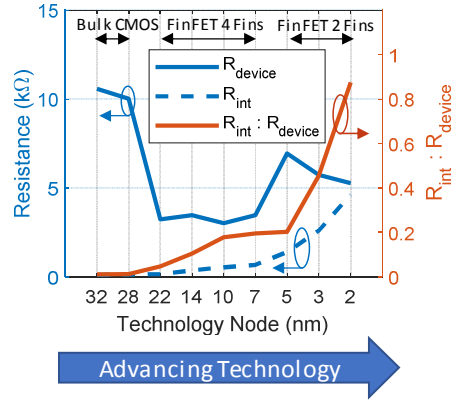


Fig 2. Interconnect and device resistance trends for advancing technology nodes

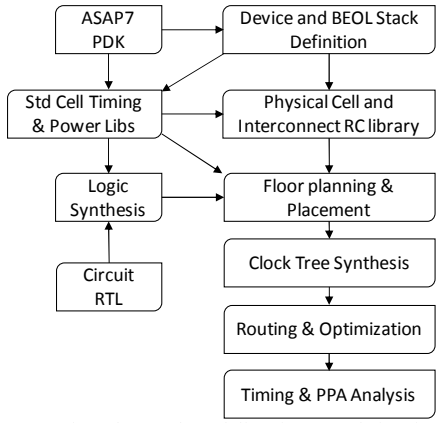


Fig. 3. Design Flow and Modeling framework for circuit-based analysis.

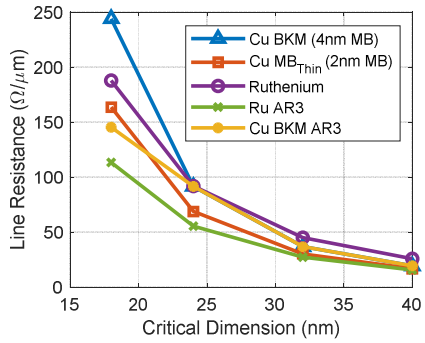


Fig 4. Line resistance for different CD

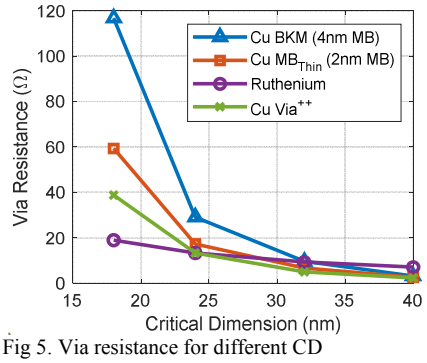


Fig 5. Via resistance for different CD

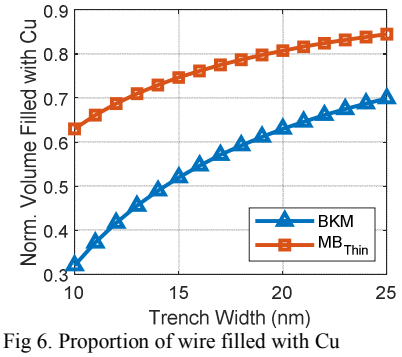


Fig 6. Proportion of wire filled with Cu

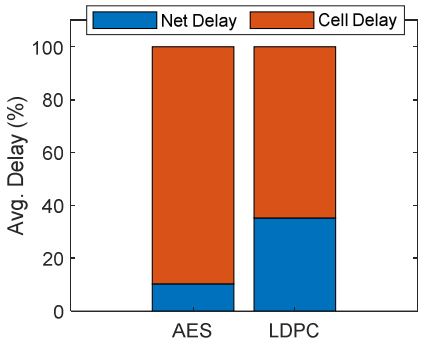


Fig 7. Percent net delay of top 50 critical paths for AES and LDPC circuits based on BKM P&R. Total net delay accounts for 10% and 35% of the total delay for the top 50 critical paths of AES and LDPC circuits, respectively.

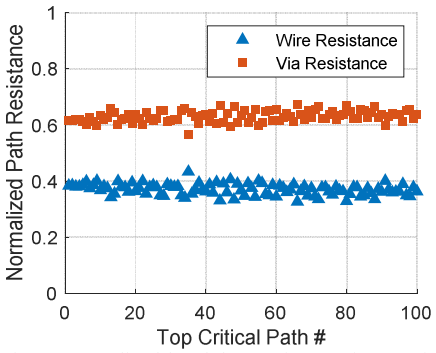


Fig 8. Normalized breakdown of net resistance from wire vs via for the top 100 critical paths. Results are based on AES P&R for BKM. Average R_{wire} is 37% for top 100 paths.

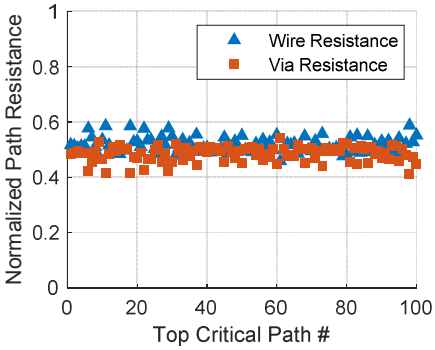


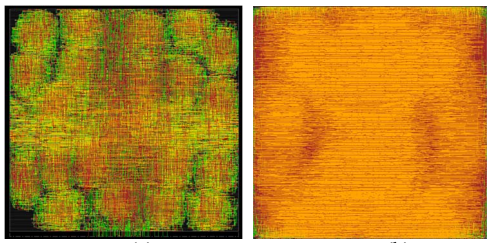
Fig 9. Normalized breakdown of net resistance due to wire vs via for the top 100 critical paths based on P&R results for BKM of LDPC circuit. Average R_{wire} is 52% for top 100 paths.

Key Parameters	Values
V_{dd} (V)	0.7
Gate length, Lg (nm)	21
Contacted Gate Pitch (nm)	54
Cell Height (tracks)	7.5
Fin Width/Pitch (nm)	6.5/27
Gate Width/Pitch (nm)	21/54
M1-M3 (Local Interconnect) CD/Pitch (nm)	18/36
M4-M5 CD/Pitch (nm)	24/48
M6 CD/Pitch (nm)	32/64

Table I. Key ASAP7 technology design Parameters

Scenario 1: BEOL MB Scaling		
Case	Wire MB t_{bottom}/t_{side} (nm)	Via MB t_{bottom}/t_{side} (nm)
BKM	4 nm / 3 nm	4 nm / 3 nm
MB_{thin}	2 nm / 1.5 nm	2 nm / 1.5 nm
Via^{++}	4 nm / 3 nm	1 nm / 1 nm
Scenario 2: Local Interconnect MB Scaling		
M_{local}-MB_{thin}	2 nm / 1.5 nm	2 nm / 1.5 nm
M_{local}-MB_{thin}(wire)	2 nm / 1.5 nm	4 nm / 3 nm
M_{local}-MB_{thin}(via)	4 nm / 3 nm	2 nm / 1.5 nm
Scenario 3: Ru and High AR Local Interconnects		
M_{local}-Ru	0.3nm (TiO_2)	0.3nm (TiO_2)
M_{local}-Ru-AR3	0.3nm (TiO_2)	0.3nm (TiO_2)
M_{local}-BKM-Cu-AR3	4 nm / 3 nm	4 nm / 3 nm

Table II. Different scenario assumptions in this work for MB and adhesion (TiO_2) thickness for bottom and sidewalls



(a) (b)

Fig. 10. BKM GDS layout for (a) AES and (b) LDPC Circuit.

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Scenario 1: BEOL MB Scaling (AES)		
Case	Eff. Freq. [GHz]	**Δ% Eff. Freq.**
BKM	2.55	
MB_{thin}	2.70	5.95%
Via^{++}	2.69	5.66%
Scenario 2: Local Interconnect MB Scaling (AES)		
M_{local}-MB_{thin}	2.67	4.81%
M_{local}-MB_{thin}(wire)	2.58	1.03%
M_{local}-MB_{thin}(via)	2.66	4.26%
Scenario 3: Ru and High AR Local Interconnects (AES)		
M_{local}-Ru	2.72	6.52%
M_{local}-Ru-AR3	2.67	4.81%
M_{local}-BKM-Cu-AR3	2.54	-0.25%

Table III. AES results for fixed P&R design

Scenario 1: BEOL MB Scaling (LDPC)		
Case	Eff. Freq. [GHz]	**Δ% Eff. Freq.**
BKM	0.99	
MB_{thin}	1.14	15.56%
Via^{++}	1.08	9.19%
Scenario 2: Local Interconnect MB Scaling (LDPC)		
M_{local}-MB_{thin}	1.081	9.19%
M_{local}-MB_{thin}(wire)	1.029	3.91%
M_{local}-MB_{thin}(via)	1.034	4.45%
Scenario 3: Ru and High AR Local Interconnects (LDPC)		
M_{local}-Ru	1.11	12.47%
M_{local}-Ru-AR3	1.09	10.50%
M_{local}-BKM-Cu-AR3	0.97	-1.85%

Table IV. LDPC results for fixed P&R design

Scenario 1: BEOL MB Scaling (AES)											
Case	Eff. Freq. [GHz]	**Δ% Eff. Freq.**	P_{tot} [mW]	P_{in} [mW]	P_{sw} [mW]	P_{lk} [mW]	PDP	**Δ% PDP**	Cell Count (#)	Buf/Inv (#)	Total WL [μm]
BKM	2.55	0	5.59	2.64	2.95	0.002	2.19		3,719	6,934	64,810
MB_{thin}	2.79	9.49%	5.96	2.82	3.14	0.002	2.14	-2.72%	3,579	6,466	63,032
Via^{++}	2.73	7.10%	5.96	2.81	3.15	0.002	2.18	-0.55%	3,620	6,502	64,185
Scenario 2: Local Interconnect MB Scaling (AES)											
M_{local}-MB_{thin}	2.69	5.66%	5.91	2.78	3.12	0.002	2.19	-0.01%	3,777	6,781	64,386
M_{local}-MB_{thin}(wire)	2.62	2.62%	5.68	2.66	3.02	0.002	2.17	-1.11%	3,859	6,756	65,900
M_{local}-MB_{thin}(via)	2.64	3.43%	5.75	2.69	3.06	0.002	2.18	-0.57%	3,629	6,787	65,450
Scenario 3: Ru and High AR Local Interconnects (AES)											
M_{local}-Ru	2.77	8.58%	6.02	2.88	3.13	0.002	2.17	-0.92%	3,487	6,355	62,329
M_{local}-Ru-AR3	2.78	8.88%	6.14	2.86	3.28	0.002	2.21	0.83%	3,513	6,070	63,776
M_{local}-BKM-Cu-AR3	2.55	0.00%	5.63	2.60	3.03	0.002	2.21	0.64%	3,706	6,667	64,876

Table V. Final Optimization Results for AES Circuit

Scenario 1: BEOL MB Scaling (LDPC)											
Case	Eff. Freq. [GHz]	**Δ% Eff. Freq.**	P_{tot} [mW]	P_{in} [mW]	P_{sw} [mW]	P_{lk} [mW]	PDP	**Δ% PDP**	Cell Count (#)	Buf/Inv (#)	Total WL [μm]
BKM	0.99	0	38.17	10.07	28.09	0.01	38.54		81,695	31,355	796,050
MB_{thin}	1.20	21.11%	43.65	11.98	31.66	0.01	36.39	-5.57%	75,758	27,733	765,692
Via^{++}	1.16	17.59%	43.13	11.35	31.77	0.01	37.04	-3.89%	80,419	30,435	792,815
Scenario 2: Local Interconnect MB Scaling (LDPC)											
M_{local}-MB_{thin}	1.20	20.82%	44.40	11.75	32.64	0.01	37.11	-3.71%	75,983	27,858	755,628
M_{local}-MB_{thin}(wire)	1.15	16.10%	44.25	11.77	32.48	0.01	38.48	-0.13%	78,863	30,027	776,213
M_{local}-MB_{thin}(via)	1.11	11.61%	41.84	11.06	30.77	0.01	37.85	-1.78%	81,111	31,251	795,236
Scenario 3: Ru and High AR Local Interconnects (LDPC)											
M_{local}-Ru	1.19	20.10%	43.58	11.48	32.10	0.01	36.64	-4.92%	77,033	27,815	768,353
M_{local}-Ru-AR3	1.19	20.10%	45.99	11.84	34.14	0.01	38.66	0.31%	74,823	27,312	746,873
M_{local}-BKM-Cu-AR3	1.02	3.06%	43.12	10.93	32.18	0.01	42.25	9.63%	82,698	33,014	801,275

Table VI. Final Optimization Results for LDPC Circuit